Module 8 — Stretch: Cross-Encoder Re-Ranking

> ***Honors Track.*** *This stretch is for learners who have completed all required Week 8 work (Reading, Drill, Lab, Integration), are On Track or Advanced, and are attending consistently. See the program's Honors Track policy for full eligibility. Stretch participation is graded but not required for program completion; it counts toward Honors distinction.*

In this stretch you will add a **cross-encoder re-ranking stage** to the lab's hybrid retriever. Cross-encoders score `(query, passage)` pairs jointly rather than independently — they produce a more discriminative ranking, but at a real latency cost. The pedagogical question is: when does the latency overhead pay off?

Due: Thursday end of day. Resubmissions are accepted through Saturday of the assignment week.

Prerequisite: Lab Challenge Tier 1 (alpha sweep) is recommended. Learners who completed Tier 1 will recognize the diminishing-returns argument that re-ranking re-opens.

What You Will Produce

This stretch is a **standalone repo** spawned from the GitHub Classroom assignment `M8-S28-cross-encoder-rerank`. New files (or completed implementations):

1. **`rerank.py`** — your `cross_encoder_rerank(query, candidates, k_out=5)` and `rerank_search(client, query, embedder, k_in=50, k_out=5)` implementations.
2. **`rerank_report.md`** — ~250-word cost/benefit writeup.
3. **PR description** — recall@5, MRR, and per-query latency for the rerank pipeline; comparison to the lab's hybrid baseline.

Setup

> ***Submission flow — all Module 8 assignments.*** *Module 8 uses a fork-and-submit flow instead of GitHub Classroom. Same pattern as the drill, lab, integration, and Tuesday stretch.*

1. **Fork the template repo to your GitHub account.** Open github.com/LevelUp-Applied-AI/m8-s28 and click **Fork** → owner = your personal GitHub account → **Create fork**.
2. **Enable Actions on your fork.** On your forked repo, click the **Actions** tab → **I understand my workflows, go ahead and enable them**.
3. **Clone *your* fork (not the upstream repo), then branch:**
```
git clone https://github.com/<your-github-username>/m8-s28.git
cd m8-s28
git checkout -b stretch-cross-encoder-rerank
```
4. Install dependencies (cached from the lab — should be fast):
```
pip install -r requirements.txt
```
5. Bring up Weaviate and ingest the corpus once with the provided helper (same as the Tuesday stretch).

Tasks

1. Implement `cross_encoder_rerank(query, candidates, k_out=5) -> list[str]`

Use `cross-encoder/ms-marco-MiniLM-L-6-v2` from `sentence-transformers`.

```
from sentence_transformers import CrossEncoder
ce = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-6-v2")
```

Score each `(query, candidate.text)` pair; sort descending; return top-`k_out` IDs.

2. Build the two-stage retriever — `rerank_search(client, query, embedder, k_in=50, k_out=5)`

Hybrid retrieve `k_in=50` (using the provided `hybrid_search` helper); cross-encoder re-rank to `k_out=5`. Return `list[str]` of `doc_id` s.

3. Measure

- recall@5 and MRR on the 60-pair labeled set (using the provided `evaluate_retriever` helper).
- Per-query latency (ms) for **each stage** — hybrid retrieval, cross-encoder re-rank, total.
- Comparison to the lab's hybrid baseline (which you can also run in this repo).

4. Cost/benefit writeup — `rerank_report.md` (~250 words)

Address:

- When does re-ranking pay off?
- What is the latency overhead per query?
- At what corpus size or query volume does it stop being worth it?

Use specific numbers from your measurements; "it depends" is not an answer.

Submission

1. Commit `rerank.py`, `rerank_report.md`, and any supporting files.
2. Push the branch (`git push origin stretch-cross-encoder-rerank`) and **open a PR *within your fork*** — base branch `main`, compare branch `stretch-cross-encoder-rerank`, **base repository = your fork** (change the dropdown from `LevelUp-Applied-AI/m8-s28` to `<your-username>/m8-s28`). If Actions are disabled on your fork, see Setup step 2 — re-enable, then `git commit --allow-empty -m "ci: trigger" && git push` to re-run CI.
3. PR description must include:
   - recall@5, MRR, and per-query latency for the rerank pipeline.
   - Comparison to the lab's hybrid baseline.
4. Paste your PR URL into TalentLMS → Module 8 → Thursday Stretch to submit this assignment.

Grading

This stretch is **rubric-graded by a TA** across 3 weighted dimensions (Two-stage retriever correctness 40%, Latency + recall measurement 30%, Cost/benefit writeup quality 30%), scored Exemplary (4) / Proficient (3) / Developing (2) / Insufficient (1). The weighted sum is scaled to 100.

The autograder verifies output shape and a behavioral recall@5 floor on a fixture subset. The TA rubric drives the final score.

Expected Outcome

A reasonable implementation typically achieves:

- recall@5 lift of ~5–10 points over the hybrid baseline on the 60-pair labeled set.
- Latency overhead ~50–200 ms/query depending on hardware (the cross-encoder scores 50 pairs sequentially per query).

The cost/benefit writeup should identify the cross-over point — at low query volume the latency is acceptable; at high QPS or large corpus sizes the cross-encoder layer becomes the bottleneck and a learned re-ranker (M9+ territory) or aggressive caching is the right next step.